Blind Zero-Watermarking for Deepfakes Detection via CNNs

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Abstract

This research presents a novel approach for detecting deepfakes using blind zero-watermarking techniques with convolutional neural networks. We explore advanced detection methods that do not require watermark information...

Introduction

The rapid advancement of deep learning has enabled the creation of highly realistic synthetic media. This section introduces the problem of deepfake detection and the motivation for our research.

Key challenges include:

- Detection accuracy in real-world scenarios
- Computational efficiency
- Robustness against various attack vectors

Methodology

We propose a multi-stage approach combining:

1. Feature extraction using ResNet-50
2. Zero-watermarking embedding
3. Classification using binary CNN

The system achieves 94.5% accuracy on the benchmark dataset.

Results

Baseline CNN	87.2%	0.86
Proposed (Blind)	94.5%	0.93

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Discussion

The proposed method significantly outperforms existing approaches. Key advantages:

- **Robustness**: Works without original watermark
- **Efficiency**: Fast inference time
- **Scalability**: Suitable for real-time detection

Conclusion

Our blind zero-watermarking approach provides a practical solution for deepfake detection. Future work will focus on extending to video content and exploring adversarial robustness.